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LiverTox: Clinical and Research Information on Drug-Induced Liver Injury [Internet]. Bethesda (MD): National Institute of Diabetes and Digestive and Kidney Diseases; 2012-.

Lingzhi, Reishi

Last Update: October 5, 2024.

OVERVIEW

Introduction

Lingzhi (*Ganoderma lingzhi*), also known as Reishi (usually *Ganoderma lucidum*), is a large, reddish-brown mushroom with a distinctive fan-like appearance that is used in traditional Chinese medicine for improving health, restoring vitality, and preventing illness. Lingzhi is well tolerated and has not been associated with serum aminotransferase elevations during therapy but has been implicated in rare single instances of acute liver injury, although in most reports other diagnoses were not adequately excluded.

Background

Lingzhi (*Ganoderma lingzhi* or *lucidum*) is a large, reddish-brown mushroom that grows at the base of trees or on tree stumps. Because it is rare in the wild, it is cultivated using logs, sawdust, or wood chips from deciduous trees. Lingzhi has a distinctive bitter taste, and is used in cooking and in making herbal tea. Its major use, however, has been in traditional Chinese medicine where it is considered an immortality herb or “divine” mushroom, capable to improving health, promoting longevity, preventing illness, and restoring youthful vigor. It has also been used as an adjunct to cancer chemotherapy to enhance antitumor activity. Known as Reishi in Japan and Lingzhi in China, it is now widely used even in Western countries where it is sold as an extract and frequently found in multiingredient dietary supplements that are purported to improve energy, vitality, and mental concentration. Studies in cell culture and animal models suggest that Lingzhi has hypoglycemic, antiinflammatory, antihypertensive, analgesic, antiviral and even antineoplastic activities. However, none of the purported beneficial activities has been demonstrated in adequately sized and properly controlled clinical trials in humans. Compounds found in Lingzhi extracts include polysaccharides, triterpenoids, nucleosides, ergosterols, fatty acids, proteins, peptides and trace elements, but the biologically active component(s) responsible for its purported effects are unknown. Various forms of Lingzhi or Reishi are available either alone as an extract or combined with other herbs and nutritional agents in multiingredient dietary products. The usual dose ranges from 500 mg to 3 grams daily. There is little information about common side effects of Lingzhi extracts in humans. In small short term clinical trials it was usually described as having few or no adverse effects.

Hepatotoxicity

In multiple small, short term placebo-controlled trials of Lingzhi or Reishi, serum aminotransferase levels did not change during treatment, and there were no reported instances of clinically apparent liver injury or jaundice. Furthermore, Lingzhi and Reishi are not listed in the many reviews of causes of liver injury from herbal and dietary supplements. Nevertheless, there have been a small number of reports from China, Japan, Thailand, and India of clinically apparent liver injury attributed to Lingzhi or Reishi or their extracts. The latency to onset was typically 1 to 2 months, but ranged from a few days to as long as 6 months, presenting with symptoms of fatigue, nausea, abdominal pain, poor appetite, dark urine, and jaundice. In

published cases, the liver injury has been hepatocellular and immune allergic reactions (fever, rash) were not common. In some instances, low levels of autoantibodies were found but immunoglobulins were typically normal and liver biopsies were more suggestive of drug induced liver injury than autoimmune hepatitis. Severity in published cases ranged from mild and asymptomatic elevations in serum aminotransferase levels to clinically apparent hepatitis and even severe hepatitis with hepatic failure. Recovery was rapid once the herbal product was stopped and complete within 1 to 3 months with or without corticosteroid therapy. There have been no reports of chronicity. Many of the cases were poorly documented and the attribution to *Ganoderma* species somewhat weak. In some cases, other common causes of liver injury were not excluded (such as gallstones, acetaminophen overdose, exposure to other herbal products, contaminants of the herbal product). In view of the extensive worldwide use of *Ganoderma lucidum* and Lingzhi, clinically apparent liver injury from its use must be extremely rare.

Likelihood score: D (possible rare cause of clinically apparent liver injury).

Mechanism of Injury

The mechanism and the ingredient in Reishi and Lingzhi responsible for the liver injury have not been identified. The most characteristic ingredients are polysaccharides and triterpenoids which have been extensively studied in animals and not been linked to liver injury.

Outcome and Management

Cases of liver injury from Reishi have ranged in severity from minimally symptomatic elevations in serum aminotransferase levels to cases of mild hepatitis resolving rapidly upon stopping to severe hepatitis with symptoms and signs of acute liver failure. Liver injury from Reishi is typically hepatocellular but self-limited in course and chronic injury has not been described. Corticosteroids have been used in some cases but with unclear benefit. Cases of rechallenge have not been reported.

Drug Class: Herbal and Dietary Supplements

Other names: Reishi, Yeongji, *Ganoderma lucidum*, *Ganoderma lingzhi*, *Ganoderma sichuanense*, Divine mushroom.

PRODUCT INFORMATION

REPRESENTATIVE TRADE NAMES

Lingzhi – Generic

DRUG CLASS

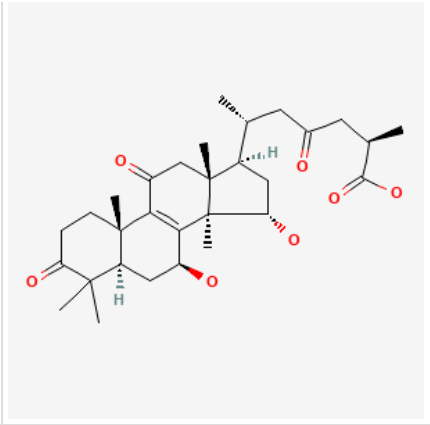
Herbal and Dietary Supplements

SUMMARY INFORMATION

Fact Sheet at MedlinePlus, NLM

CHEMICAL FORMULA AND STRUCTURE

DRUG	CAS REGISTRY NUMBER	MOLECULAR FORMULA	STRUCTURE
Ganoderic Acid A	<u>81907-62-2</u>	C30-H44-O7	



Beta-Glucan	9012-72-0	C18-H32-O16	Not Available
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ANNOTATED BIBLIOGRAPHY

References updated: 05 October 2024

Abbreviations: DILI, drug induced liver injury; HDS, herbal and dietary supplements.

(Expert review of hepatotoxicity published in 1999; several herbal medications are discussed, but not Reishi or Lingzhi).

Zimmerman HJ. Unconventional drugs. Miscellaneous drugs and diagnostic chemicals. In, Zimmerman, HJ. Hepatotoxicity: the adverse effects of drugs and other chemicals on the liver. 2nd ed. Philadelphia: Lippincott, 1999: pp. 731-4.

Yuen MF, Ip P, Ng WK, Lai CL. Hepatotoxicity due to a formulation of Ganoderma lucidum (lingzhi). J Hepatol. 2004;41:686-7. [PubMed: 15464254]
(78 year old Chinese woman who had taken Lingzhi for a year and then switched to a powdered formulation developed fatigue and dark urine one month later [bilirubin 15.8 mg/dL, ALT 306, Alk P 388 U/L, INR normal], improved slowly after stopping and had near normal liver tests 5 months later).

Wachtel-Galor S, Tomlinson B, Benzie IF. Ganoderma lucidum ("Lingzhi"), a Chinese medicinal mushroom: biomarker responses in a controlled human supplementation study. Br J Nutr. 2004;91:263-9. [PubMed: 14756912]
(Among 18 healthy Chinese adults treated with Lingzhi [1.44 gm] or placebo daily for 28 days followed after a 4-6 week washout period with crossover to the alternate treatment, serum markers of antioxidant, vitamin, and lipid status did not change significantly in either group, and therapy was well tolerated with no change in blood counts or serum creatinine, ALT and AST levels).

Tang W, Gao Y, Chen G, Gao H, Dai X, Ye J, Chan E, Huang M, Zhou S. A randomized, double-blind and placebo-controlled study of a Ganoderma lucidum polysaccharide extract in neurasthenia. J Med Food. 2005;8:53-8. [PubMed: 15857210]
(Among 132 Chinese patients with neurasthenia treated with a polysaccharide extract of Ganoderma lucidum [1800 mg] or placebo capsules 3 times daily for 8 weeks, wellness and fatigue scores improved more with Lingzhi than placebo and therapy was well tolerated and "not associated with hematologic, liver, renal or biochemical toxicity").

Wicks SM, Tong R, Wang CZ, O'Connor M, Karrison T, Li S, Moss J, Yuan CS. Safety and tolerability of Ganoderma lucidum in healthy subjects: a double-blind randomized placebo-controlled trial. Am J Chin Med. 2007;35:407-14. [PubMed: 17597499]
(Among 16 volunteers treated with Lingzhi extract [2 gm] or placebo twice daily for 10 days, there were no significant changes in lymphocyte subsets or in ALT, AST, Alk P or bilirubin levels and "no adverse effects of clinical importance").

- Wanmuang H, Leopairut J, Kositchaiwat C, Wananukul W, Bunyaratvej S. Fatal fulminant hepatitis associated with *Ganoderma lucidum* (Lingzhi) mushroom powder. *J Med Assoc Thai*. 2007;90:179-81. [PubMed: 17621752]
(47 year old Thai woman developed jaundice and coma 2 months after switching from boiled Lingzhi slices to a powdered form of 200 mg daily [bilirubin 40.4 mg/dL, ALT 276 U/L, AST 280 U/L, Alk P 132 U/L, INR 4.7], with rapid deterioration and death 6 days later; autopsy showing multilobular necrosis).
- Noguchi M, Kakuma T, Tomiyasu K, Kurita Y, Kukihara H, Konishi F, Kumamoto S, et al. Effect of an extract of *Ganoderma lucidum* in men with lower urinary tract symptoms: a double-blind, placebo-controlled randomized and dose-ranging study. *Asian J Androl*. 2008;10:651-8. [PubMed: 18097503]
(Among 50 Chinese male adults with prostate related symptoms treated with Lingzhi [0.6, 6, or 60 mg] or placebo once daily for 8 weeks, symptoms improved slightly with the 6 mg dose but there was no change in prostate size or residual urinary volume, while adverse events were uncommon and there was "no treatment related hematologic, hepatic, or renal toxicity").
- Noguchi M, Kakuma T, Tomiyasu K, Yamada A, Itoh K, Konishi F, Kumamoto S, Shimizu K, Kondo R, Matsuoka K. Randomized clinical trial of an ethanol extract of *Ganoderma lucidum* in men with lower urinary tract symptoms. *Asian J Androl*. 2008;10:777-85. [PubMed: 18097505]
(Among 88 men with lower urinary tract symptoms treated with Ganoderma lucidum [6 mg] or placebo once daily for 12 weeks, there were no changes in urinary symptoms, prostate size, or residual urinary volume and there was "no treatment related hematologic, hepatic or renal toxicity").
- Chu TT, Benzie IF, Lam CW, Fok BS, Lee KK, Tomlinson B. Study of potential cardioprotective effects of *Ganoderma lucidum* (Lingzhi): results of a controlled human intervention trial. *Br J Nutr*. 2012;107:1017-27. [PubMed: 21801467]
(Among 26 patients with cardiovascular risk factors treated with Lingzhi [1.44 gm] or placebo daily for 12 weeks in a double-blind cross-over design, body weight, lipids, blood pressure, and fasting glucose levels did not change, while a slight improvement occurred with insulin levels and there were "no adverse effects in laboratory safety parameters").
- Soares AA, de Sá-Nakanishi AB, Bracht A, da Costa SM, Koehnlein EA, de Souza CG, Peralta RM. Hepatoprotective effects of mushrooms. *Molecules*. 2013;18:7609-30. [PMC free article: PMC6270077] [PubMed: 23884116]
(Review of the hepatoprotective activity of mushrooms, focused largely on in vivo studies of Ganoderma lucidum effects in animal models of carbon tetrachloride or galactosamine hepatotoxicity suggesting that the polysaccharides and triterpenoids are the active principles, and stressing the need for clinical trials in humans with acute liver failure).
- Klupp NL, Kiat H, Bensoussan A, Steiner GZ, Chang DH. A double-blind, randomised, placebo-controlled trial of *Ganoderma lucidum* for the treatment of cardiovascular risk factors of metabolic syndrome. *Sci Rep*. 2016;6:29540. [PMC free article: PMC4980683] [PubMed: 27511742]
(Among 84 Australian adults with diabetes and the metabolic syndrome treated with Ganoderma lucidum alone [3 gm] or with Cordyceps [amount not given], or placebo for 16 weeks, there were no significant changes in body weight, blood pressure, or in fasting blood glucose, HbA1c, and serum lipid levels in the 3 groups; side effects were mild and similar to those in patients on placebo).
- Teschke R, Wolff A, Frenzel C, Schulze J, Eickhoff A. Herbal hepatotoxicity: a tabular compilation of reported cases. *Liver Int* 2012; 32: 1543-56. [PubMed: 22928722]
(A systematic compilation of all publications on the hepatotoxicity of specific herbal

- products identified 185 publications on 60 different herbs, herbal drugs and supplements but does not list or mention Reishi or Lingzhi).*
- Navarro VJ, Seeff LB. Liver injury induced by herbal complementary and alternative medicine. Clin Liver Dis 2013; 17: 715-35. [PubMed: 24099027]
(Review of the epidemiology, regulatory status, diagnosis, pathogenesis and causes of liver injury from herbal products with specific discussion of conjugated linoleic acid, ephedra, germander, green tea, usnic acid, flavocoxid, aloe vera, chaparral, greater celandine, black cohosh, comfrey, kava, skullcap, valerian, noni juice, pennyroyal and traditional herbal remedies; no mention of Reishi or Lingzhi).
- Navarro VJ, Barnhart H, Bonkovsky HL, Davern T, Fontana RJ, Grant L, Rledy KR, et al. Liver injury from herbals and dietary supplements in the U.S. Drug-Induced Liver Injury Network. Hepatology 2014; 60: 1399-408. [PMC free article: PMC4293199] [PubMed: 25043597]
(Among 839 cases of liver injury from drugs collected in the US between 2004 and 2013, 130 were due to HDS products, including 45 from body building agents [probably anabolic steroids] and 85 from diverse HDS products but no case was attributed specifically to Reishi or Lingzhi).
- Brown AC. Liver toxicity related to herbs and dietary supplements: Online table of case reports. Part 2 of 5 series. Food Chem Toxicol 2017; 107: 472-501. [PubMed: 27402097]
(Description of an online compendium of cases of liver toxicity attributed to HDS products, does not list or discuss Reishi or Lingzhi).
- Medina-Caliz I, Garcia-Cortes M, Gonzalez-Jimenez A, Cabello MR, Robles-Diaz M, Sanabria-Cabrera J, Sanjuan-Jimenez R, et al.; Spanish DILI Registry. Herbal and dietary supplement-induced liver injuries in the Spanish DILI Registry. Clin Gastroenterol Hepatol. 2018;16:1495-1502. [PubMed: 29307848]
(Among 856 cases of hepatotoxicity enrolled in the Spanish Drug-Induced Liver Injury Registry between 1994 and 2016, 32 were attributed to herbal products, the most frequent cause being green tea [n=8] and Herbalife products [n=6], no mention of Reishi or Lingzhi).
- Wang GH, Wang LH, Wang C, Qin LH. Spore powder of Ganoderma lucidum for the treatment of Alzheimer disease: A pilot study. Medicine (Baltimore). 2018;97:e0636. [PMC free article: PMC5959386] [PubMed: 29742702]
(Among 42 Chinese patients with Alzheimer disease treated with spore powder of Lingzhi [1000 mg] or placebo 3 times daily for 6 weeks, there were no differences in changes in neuropsychic symptoms or quality of life measures and no differences in adverse events between the two groups; no mention of ALT or AST levels).
- Wang GH, Li X, Cao WH, Li J, Wang LH. A retrospective study of Ganoderma lucidum spore powder for patients with epilepsy. Medicine (Baltimore). 2018;97:e10941. [PMC free article: PMC5999473] [PubMed: 29879039]
(Among 18 Chinese patients with epilepsy treated with spore powder of Lingzhi [1000 mg] 3 times daily for 8 weeks, there were no significant changes in seizure frequency or quality of life measures and adverse events were generally mild; no mention of ALT levels or hepatotoxicity).
- Qiu Z, Zhong D, Yang B. Preventive and therapeutic effect of Ganoderma (Lingzhi) on liver injury. Adv Exp Med Biol. 2019;1182:217-242. [PubMed: 31777021]
(Review of studies of the beneficial effects of Lingzhi on liver injury).
- Liu J, Mao JJ, Li SQ, Lin H. Preliminary efficacy and safety of Reishi & Privet Formula on quality of life among non-small cell lung cancer patients undergoing chemotherapy: a randomized placebo-controlled trial. Integr Cancer Ther. 2020;19:1534735420944491. [PMC free article: PMC7450289] [PubMed: 32840126]

(Among 82 Chinese adults with lung cancer treated with an oral herbal product with Reishi and Privet vs placebo for 6 weeks, there were no significant differences in changes in quality of life scores or in liver related adverse events between the two groups).

Ahmad R, Riaz M, Khan A, Aljamea A, Algheryafi M, Sewaket D, Alqathama A. *Ganoderma lucidum* (Reishi) an edible mushroom; a comprehensive and critical review of its nutritional, cosmeceutical, mycochemical, pharmacological, clinical, and toxicological properties. *Phytother Res*. 2021;35:6030-6062. [PubMed: 34411377]

(Review of the safety and efficacy of Ganoderma lucidum from the published literature concludes that adequately powered clinical studies are needed to verify the biologic activities of Reishi that have been shown in laboratory and animal studies; mentions that hepatotoxicity has been reported in humans with its use [Wanmuang 2007]).

Kogure T, Koiwai A, Fukushima D, Satoh M, Murakami K, Hirota M, Endo K, et al. Hypereosinophilia with hepatic nodule formation caused by *Ganoderma lucidum*. *Intern Med*. 2021;60:3897-3903. [PMC free article: PMC8758446] [PubMed: 34911873]

(61 year old Japanese man was found to have mildly abnormal liver tests 6 months after resection of colon cancer leading to discontinuation of his adjunctive cancer chemotherapy, whereupon he started a commercial herbal product containing Reishi and developed nausea and fatigue two months later and was found to have multiple hepatic nodules [ALT 66, AST 45 U/L, Alk P 445 U/L, eosinophils 14,310/ μ L], which on biopsy showed nodules with intense hypereosinophilia; stopping Reishi was followed by resolution of symptoms, abnormal liver tests and disappearance of hepatic nodules).

Ballotin VR, Bigarella LG, Brandão ABM, Balbinot RA, Balbinot SS, Soldera J. Herb-induced liver injury: systematic review and meta-analysis. *World J Clin Cases*. 2021;9:5490-5513. [PMC free article: PMC8281430] [PubMed: 34307603]

(Systematic review of the literature on HDS induced liver injury identified 446 references describing 936 cases due to 79 different herbal products, the most common being He Shou Wu [91], green tea [90] Herbalife products [64], kava kava [62] and greater celandine [48]; Lingzhi is not listed or discussed).

Bessone F, García-Cortés M, Medina-Caliz I, Hernandez N, Parana R, Mendizabal M, Schinoni MI, et al. Herbal and dietary supplements-induced liver injury in Latin America: experience from the LATINDILI Network. *Clin Gastroenterol Hepatol*. 2022;20:e548-e563. [PubMed: 33434654]

(Among 367 cases of hepatotoxicity enrolled in the Latin American Drug-Induced Liver Injury Network between 2011 and 2019, 29 [8%] were attributed to herbal products, the most frequent being green tea [n=7], Herbalife products [n=5] and garcinia [n=3]; Lingzhi is not mentioned).

Liu C, Song X, Li Y, Ding C, Li X, Dan L, Xu H, Zhang D. A comprehensive review on the chemical composition, pharmacology and clinical applications of *Ganoderma*. *Am J Chin Med*. 2023;51:1983-2040. [PubMed: 37903715]

(Review of the chemical analyses, pharmacology and clinical uses of Ganoderma species used in Chinese traditional medicine mentions that Lingzhi has more than 470 natural components including terpenoids, steroids, alkaloids, and phenols, and has antitumor, antiinflammatory, immunomodulatory, hypolipidemic, and hypoglycemic activity, and is used in China to treat malignant tumors and diabetic nephropathy).

Chen SN, Nan FH, Liu MW, Yang MF, Chang YC, Chen S. Evaluation of immune modulation by β -1,3; 1,6 D-glucan derived from *Ganoderma lucidum* in healthy adult volunteers, a randomized controlled trial. *Foods*. 2023;12:659. [PMC free article: PMC9914031] [PubMed: 36766186]

(Among 157 healthy adult volunteers treated with polysaccharides [β -glucan] derived from Ganoderma lucidum vs placebo for 12 weeks, treated patients had increases in CD3,

CD4 and CD8 lymphocytes and NK cells, but caused no serious adverse events or changes in serum ALT or AST levels).

Khan H, Reyes JVM, Seen T, Irefej B, Ahmad S. Herbal supplement-induced liver injury: a case report. *Cureus*. 2023;15:e33663. [PMC free article: [PMC9928137](#)] [PubMed: [36819353](#)]

(45 year old woman developed epigastric pain 3 days after starting an herbal tea [bilirubin 1.9 mg/dL, ALT 940 U/L, AST 1465 U/L, Alk P 145 U/L], imaging demonstrating gallstone and biliary sludge but no evidence of biliary obstruction, with rapid improvement after stopping the tea which analysis showed contained multiple herbs including Reishi, Siberian ginseng, and aloe vera).

Guedikian R, Kim B, Singh G, Alexander R. Ganoderma lingzhi (Reishi Mushroom)-induced acute liver injury in the setting of alcohol use: a case report and review of the literature. *Cureus*. 2023;15:e45953. [PMC free article: [PMC10599861](#)] [PubMed: [37885515](#)]

(47 year old man developed nausea, headache, and abdominal and leg pain which he treated with alcohol [750 mL daily] and several teaspoons of Reishi mushrooms daily for three days and then presenting to an emergency room with bilirubin 1.5 mg/dL, ALT 990 U/L, Alk P 301 U/L, which resolved rapidly with infusions of N-acetylcysteine and hydration, ALT falling to normal within 2 weeks).

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